

Segment Tree

Concepts & Qns...



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codestorywithmik



CSwithMIK

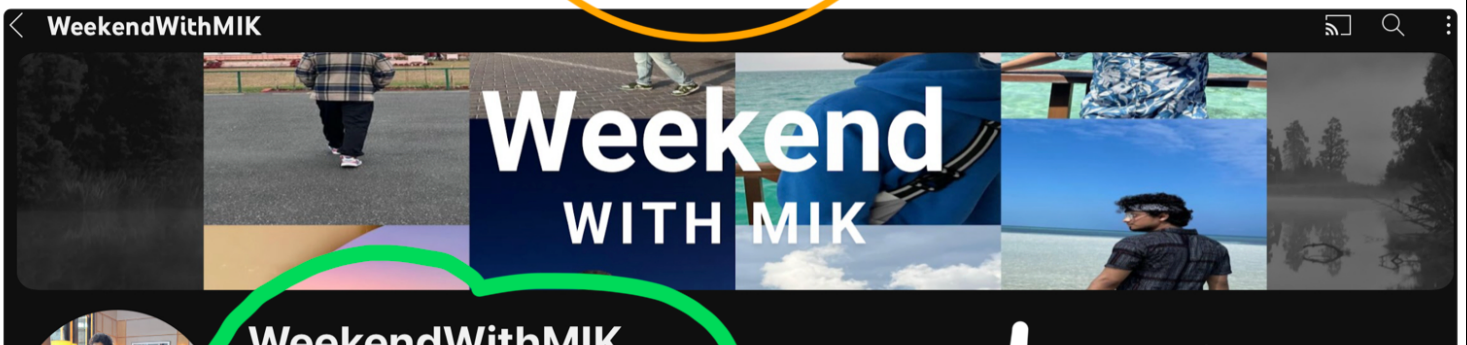


codestorywithMIK

video - 16

"No more fear of Segment Tree"

Leetcode
- 3161
Hard





WEEKENDWITHMIK

@WeekendWithMIK

Welcome to WeekendWithMIK 🌟 >

MIK

Try this channel to see
my "Life behind the scenes + Tech News"

Motivation:-

You don't need to do big things at once.

Just waking up, doing things consistently

in small steps every day is
more than enough to bring a big
change in your life...



MIK...

3161. Block Placement Queries

Hard

Topics

Companies

Hint

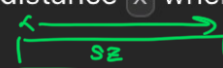
There exists an infinite number line, with its origin at 0 and extending towards the **positive** x-axis.

You are given a 2D array `queries`, which contains two types of queries:

1. For a query of type 1, `queries[i] = [1, x]`. Build an obstacle at distance `x` from the origin. It is guaranteed that there is **no** obstacle at distance `x` when the query is asked.
2. For a query of type 2, `queries[i] = [2, x, sz]`. Check if it is possible to place a block of size `sz` anywhere in the range `[0, x]` on the line, such that the block **entirely** lies in the range `[0, x]`. A block **cannot** be placed if it intersects with any obstacle, but it may touch it. Note that you do not actually place the block. Queries are separate.

$\Rightarrow (2, 4, 2)$

$\Rightarrow (2, 4, 1)$

$\Rightarrow$ 

Return a boolean array `results`, where `results[i]` is `true` if you can place the block specified in the `ith` query of type 2, and `false` otherwise.

Example 1:

Input: queries = [[1,2],[2,3,3],[2,3,1],[2,2,2]]

Output: [false,true,true]

Explanation:



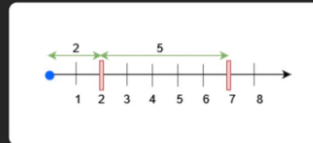
For query 0, place an obstacle at `x = 2`. A block of size at most 2 can be placed before `x = 3`.

Example 2:

Input: queries = [[1,7],[2,7,6],[1,2],[2,7,5],[2,7,8]]

Output: [true,true,false]

Explanation:



Place an obstacle at `x = 7` for query 0. A block of size at most 7 can be placed before `x = 7`.

Place an obstacle at `x = 2` for query 2. Now, a block of size at most 5 can be placed before `x = 7`, and a block of size at most 2 before `x = 2`.

Constraints:

- $1 \leq \text{queries.length} \leq 15 \cdot 10^4$
- $2 \leq \text{queries}[i].\text{length} \leq 3$ ← $\{1, x\}, \{2, x, sz\}$
- $1 \leq \text{queries}[i][0] \leq 2$
- $1 \leq x, sz \leq \min(5 \cdot 10^4, 3 \cdot \text{queries.length})$ ← $\min(5 \cdot 10^4, 45 \cdot 10^4)$
- The input is generated such that for queries of type 1, no obstacle exists at distance `x` when the query is asked.
- The input is generated such that there is at least one query of type 2.

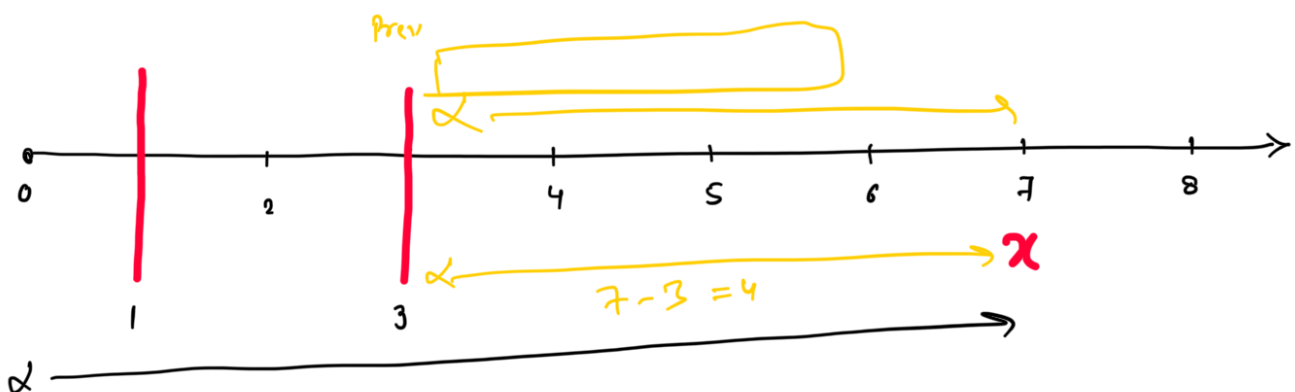
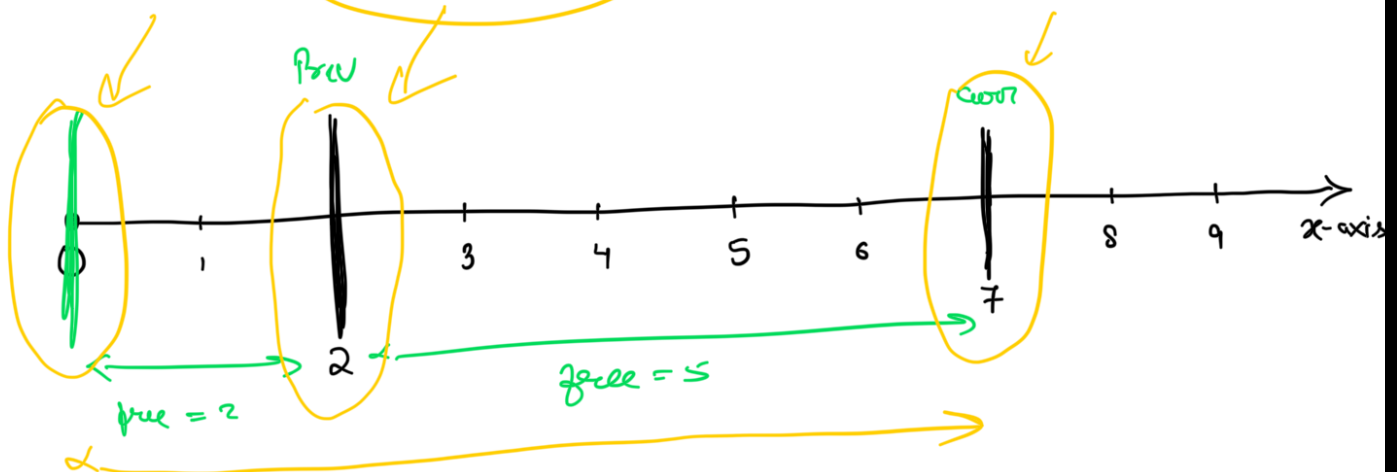
Block Force

Thought Process

queries = $\left[(1, 7), (2, 7, 6), (1, 2), (2, 7, 5), (2, 7, 6) \right]$

Set = $\{0, 2, 7\}$

$s2 = 5$
Time SZ



$$\text{query}[i] = (2, 7, 3)$$

$\uparrow$ $\uparrow$ $\uparrow$
 type x sz=3

$$(x - \text{prev}) \geq \text{sz}$$

$\Rightarrow O * (\text{obstacles iter}) \rightarrow \text{TLE}$

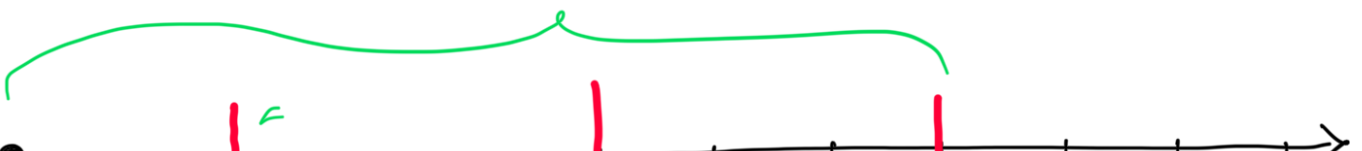
How to improve?

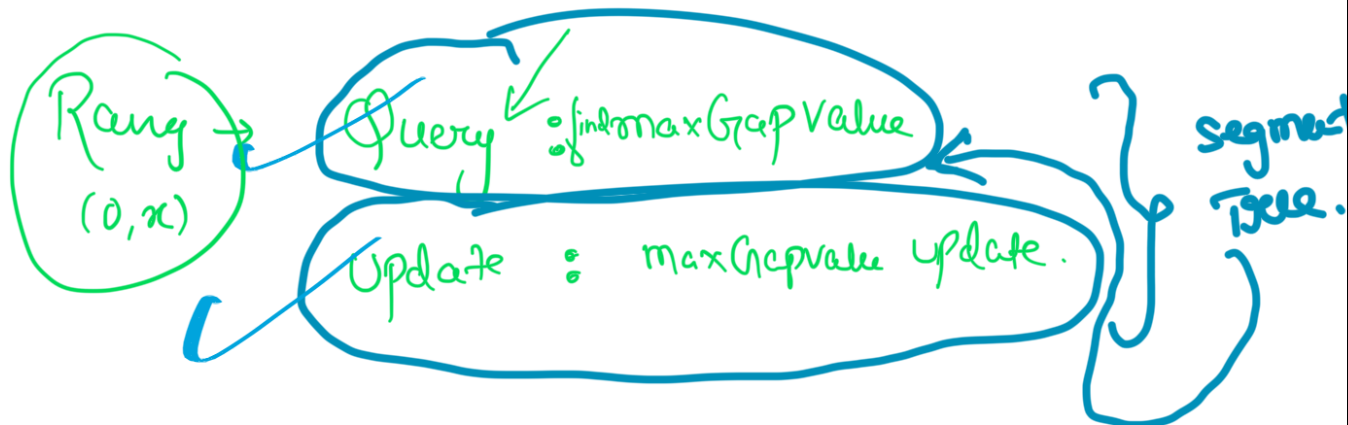
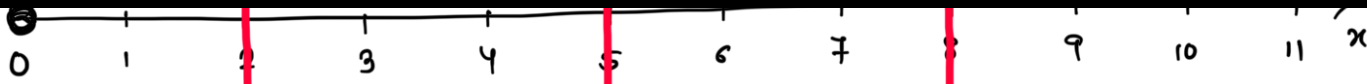
$$\text{query}[i] = (2, 8, 3)$$

$\uparrow$ $\uparrow$ $\uparrow$
 x sz

$$\text{Range} = (0, 8)$$

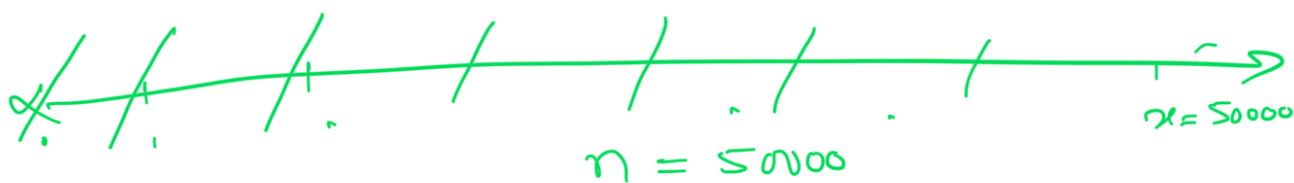
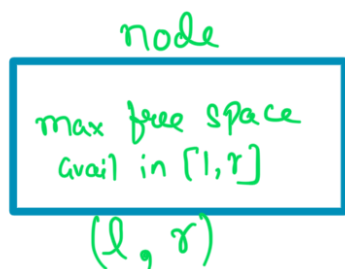
$$\text{maxGapValue} = 3$$





Segment Tree

what does a node in segment tree will represent ???



Constraints:

- $1 \leq \text{queries.length} \leq 15 * 10^4$

seg(4 * n)

- `2 <= queries[i].length <= 3`

- `1 <= queries[i][0] <= 2`

50000

- `1 <= x, sz <= min(5 * 104, 3 * queries.length)`

- The input is generated such that for queries of type 1, no obstacle exists at distance `x` when the query is asked.

- The input is generated such that there is at least one query of type 2.

Construct segment Tree.

`n = 50000;`

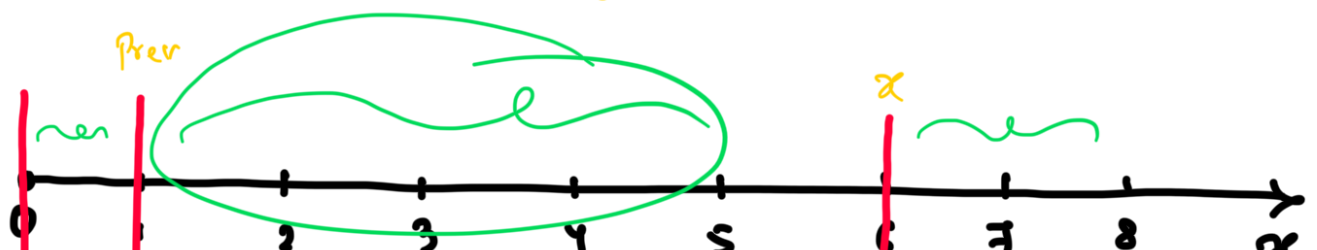
`vector<int> segTree(4*n, 0)`

Query Type-1 :

$(1, x) \rightarrow$ put an obstacle at x

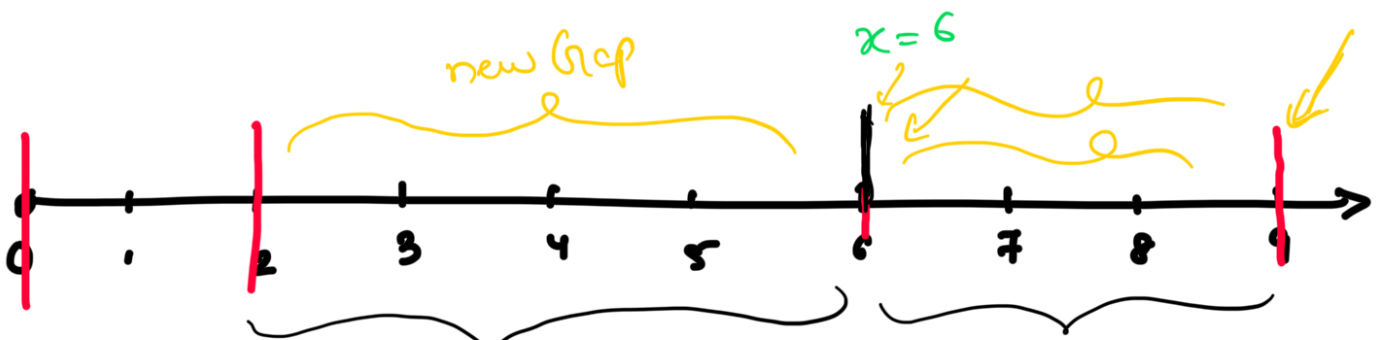
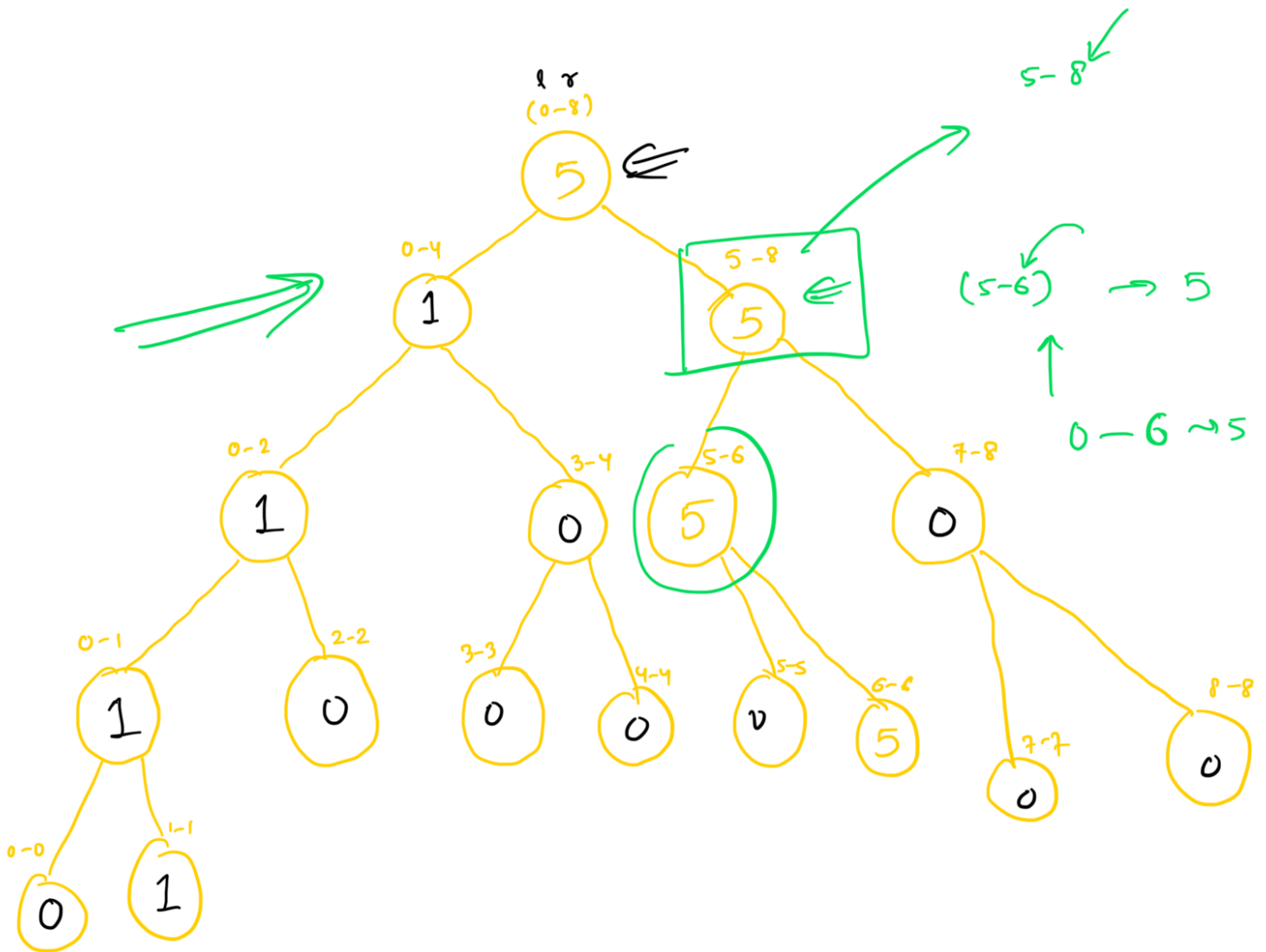
set = {0, 1, 6}

segmentTree = $x = x-1 = 6-1 = 5$



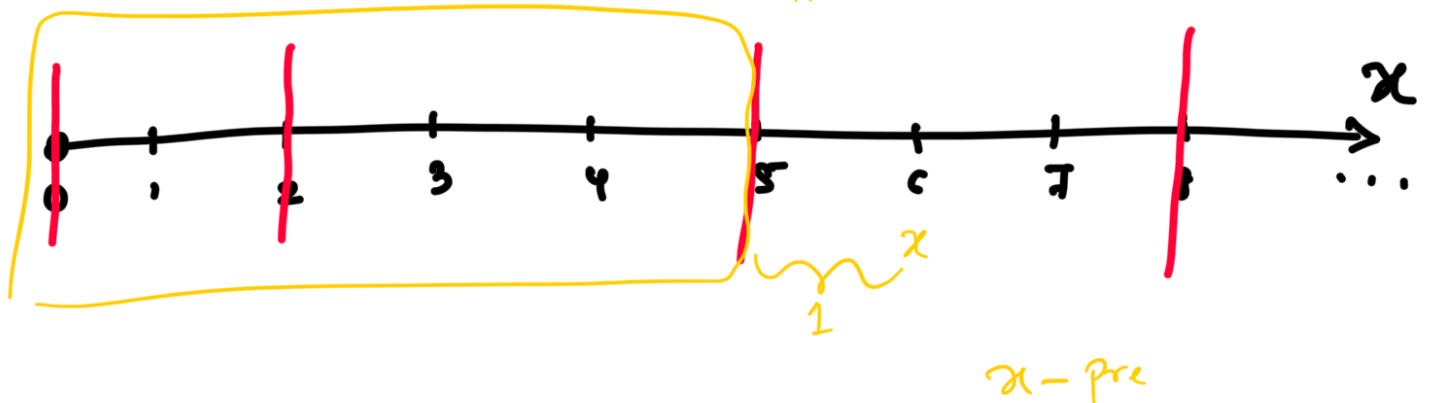
~ 5 x next

0	1	2	3	4	5	6	7	8
0	(1)	0	0	0	0	0	0	0



5

3

 $x = 6$ $set = \{0, 2, 9\}$ $it = set.upperbound(x);$ $\downarrow$ $\checkmark next = *it; // 9$ $set = \{0, 2, 6, 9\}$ $\checkmark prev = *prev(it); // 2$ seg tree
concept $updateSegTree(x, x - prev, 0, 0, n-1); \checkmark$ $updateSegTree(next, next - x, 0, 0, n-1); \checkmark$ $st.insert(x);$ Query Type = 2 $\rightarrow (2, x, sz)$ $set = \{0, 2, 5, 8\}$ it $query = (2, 6, (2))$
 sz 

it = set.upperbound(x); // x = 6

root
O

pre = *prev(it); // 5

maxGap = querySeq(0, pre, 0, 0, n-1);

temp = max(maxGap, x - pre);

res.push(temp);

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